

Semantic Ontologies in Economics

Special Issue Proposal

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1 The Challenge: AI and the Meaning of Economic Models

When economists compute a model, its precise *meaning* is distributed, much of it implicitly, across prose, notation, calibration, and code. The paper says “we solve the following model”, but its equations underdetermine both what its code computes and what the computations mean. Only an informed reader can fill the gaps, one who knows the surrounding literature and has absorbed, through their professional networks, the conventions and traditions the paper leaves unstated. The same gaps confront anyone who attempts to interpret the model’s findings, rebuild the model, or extend it. As a result, without a concrete statement of the model’s meaning to check against, computational results are hard to cross-verify, and interoperability between implementations becomes difficult.

Relying on implicit professional convention alone is costly even amongst humans, but it fails completely when AI is used to write, modify, and translate modeling code. To interpret or verify AI output, we need a statement, independent of the code, of which economic objects are computed and which relations among them are enforced; these statements are called *semantic ontologies*. Outside economics, the use of semantic ontologies has grown rapidly, because without them, AI outputs remain probabilistic, and prone to error and misunderstanding (Grüninger and Menzel 2003; Modelica Association 2023; Gleeson et al. 2010; AWS Database Blog 2026; see the *Semantic Ontologies* section of the annex). With semantic ontologies, translating models between toolkits, and even generating new modeling research, become operations a language model can carry out and be checked on cheaply, rather than manual recoding.

2 The Proposed Special Issue

We propose a special issue that constructs semantic ontologies for the domains of computational economics: domain-specific modeling languages (for example, Dynare), toolkits (for example, HARK and SSJ), empirical methods (for example, structural modeling and estimation), and model classes (for example, life-cycle, real-business-cycle, and agent-based models). Each paper takes one domain of study, proposes a semantic ontology, and demonstrates

it on worked examples of the team’s own choosing. In addition to the semantic ontologies, the working group will author a paper and a community resource of canonical models and their ontologies.¹

The special issue will also include related contributions by the working group that use the stated ontologies to compare different modeling strategies for the same problem, for instance agent-based models against heterogeneous-agent models.

The SCE working group will submit a subset of the papers, an open call will invite the rest, and a comparison paper, written jointly by the participating teams, will close the issue.

The semantic ontologies we develop will be kept in an open-source [GitHub repository](#), to which members of the community can contribute improvements and modifications.

3 Organization

Papers pass an internal review within the working group [process TBC] and then the journal’s ordinary external refereeing. We expect six to eight papers, with a session at the Society’s conference between submission and revision.

3.1 Teams and Editors

The working group’s members include developers of Dynare, HARK, and the VFI Toolkit. Committed teams: [to be listed; only teams that have agreed in writing]. The working group’s papers will come from its subgroups. The guest editors (a lead editor and at least one co-editor, neither submitting to the issue) are drawn from outside the working group [names, institutions; TBC]. Papers are submitted through the journal’s editorial system, marked for this special issue, and are refereed under the journal’s ordinary standards; the guest editors handle every paper, including the comparison paper, subject to the journal’s final editorial authority.

3.2 Timeline

Calendar dates TBC on acceptance.

- Call for papers [on acceptance]
- Submissions [+9 months]
- Internal review completed [+12 months; process TBC]
- Referee reports [+15 months]
- Revisions and the comparison paper [+24 months]

¹The nearest precedent in economics, the Macroeconomic Model Data Base (Wieland et al. 2012), compares models under common variables, common shocks, and a menu of common policy rules while each model keeps its own equations; it standardizes comparison, not meaning.

References

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